Source: Electromagnetic Composites Handbook: Models, Measurement, and Characterization, 2nd Edition
ISBN: 9781259585043
Authors: Rick Moore

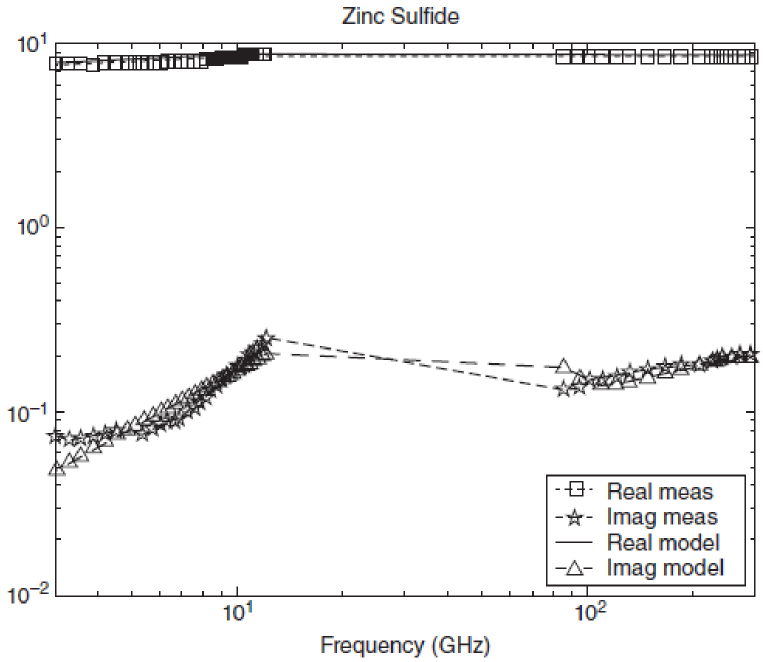
12.6. SOLID SEMICONDUCTORS VERSUS FREQUENCY

Table 12.6 shows the parametric fits of permittivity for various semiconductors. Test samples were cut from large wafers for testing. Free space measurements of the full wafer were used for mmW measurements. The functional forms are shown at the beginning of the table. Functions are those causal functions that have been derived in Chaps. 2 and 3 for dielectric and magnetic media. Other data are shown when available. These include density, measured frequency span serving as basis for fits, tabulated information, and chemical and/or physical composition. In the tables exponentials are often shown as bE—which means b*10^{-a}. Thus, –2E-004–7.6E-003j is equal to –0.0002–j0.0076. **Figure 12.9** shows a typical broadband fit. Please note that the Silicon and Germanium data were acquired over limited bandwidth.

Table 12.6 *Semiconductors*

Sample ID $\varepsilon(f) = B + Real(C) f^D + Imag(C) f^E + F(1-(f/G)^2 - j2f/H)^{-1}$						
B	C	D	E	F	G	H
Zinc sulfide; 4.09 g/cc (3–240 GHz) (nominal ε_r = 8.5 at 2 MHz; Korth Kristalle GMBH)						
2.44–6E-003j	2.58+6E-003j	0.15+6E-003j	0.15+6E-003j	1.9+3E-003j	3.5+53.24j	18.9+88.9j
Zinc selenide, 5.27 g/cc (5–240 GHz) (nominal ε_r = 8.98; Korth Kristalle GMBH)						
2.37–0.13j	2.65–4E-002j	0.15+7E-002j	0.14+6E-002j	2.53+6E-002j	136–138j	–176+42.4j
Gallium arsenide (3–240 GHz) (nominal ε_r = 8.35–11.36; US pat. 6683510)						
3.63–0.13j	3.64–7E-002j	8E-002+3E-002j	0.19–0.3j	3.73–4E-002j	76.9–367j	–547+598j
Silicon (2.33 g/cc) (9–12.4 GHz) (nominal ε_r = 10.2; Accumet Engineering)						
3.89+8E-003j	3.89+8E-003j	–2E-003+2E-003j	–2E-003+2E-003j	3.89+8E-003j	440–1702j	–7627+10622j
Germanium (5.33 g/cc) (3–10 GHz) (nominal ε_r = 16; Virginia Semiconductor)						
11.97+2.83j	13.23+3.4j	–2E-002+0.25j	0.53+1.44j	14.53+1.8j	1.1+6j	0.92+8.12j

Figure 12.9 *Measurement and model fit for Zinc Sulfide*



Disclaimer: Data are results obtained by the author and colleagues on a small number of samples purchased or obtained as free samples from suppliers. Data are not "guaranteed" by the manufacturer or author. Variations in properties may be observed and users of the products should verify properties. All material properties are guides for engineering design. Materials prepared in the laboratory should be considered as experimental and also guides for engineering design.